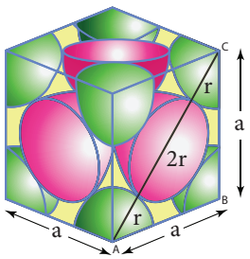
The cubic close packing is based on the face centered cubic unit cell. Let us calculate the packing efficiency in fcc unit cell.



From the figure

$AC = 4r$

$4r = a\sqrt{2}$

$r = \frac{a\sqrt{2}}{4}$

In ΔABC

$AC^2 = AB^2 + BC^2$

$AC = \sqrt{AB^2 + BC^2}$

$AC = \sqrt{a^2 + a^2} = \sqrt{2a^2} = \sqrt{2}\ a$

Volume of the sphere with radius r is

$$= \frac{4}{3}\pi\left(\frac{\sqrt{2}a}{4}\right)^3$$

$$= \frac{4}{3}\pi\left(\frac{2\sqrt{2}a^3}{64}\right)$$

$$= \frac{\sqrt{2}\ \pi a^3}{24}$$

Total number of spheres belongs to a single fcc unit cell is 4

$\therefore$ the volume of all spheres in a fcc unit cell $= 4 \times \left(\frac{\sqrt{2}\ \pi a^3}{24}\right)$

$$= \left(\frac{\sqrt{2}\ \pi a^3}{6}\right)$$

$$\text{packing efficiency} = \frac{\left(\frac{\sqrt{2}\ \pi\ a^3}{6}\right)}{\left(a^3\right)} \times 100$$

$$= \frac{\sqrt{2}\ \pi}{6} \times 100$$

$$= \frac{1.414 \times 3.14 \times 100}{6}$$

$= 74\%$

Radius ratio:

The structure of an ionic compound depends upon the stoichiometry and the size of the ions.generally in ionic crystals the bigger anions are present in the close packed arrangements and the cations occupy the voids. The ratio of radius of cation and anion $\left(\frac{r_{C^+}}{r_{A^-}}\right)$ plays an important role in determining the structure. The following table shows the rleation between the radius ratio and the structural arrangement in ionic solids.

$\left(\frac{r_{C^+}}{r_{A^-}}\right)$	Coordination number	Structure	Example
0.155 – 0.225	3	Trigonal planar	B_2O_3
0.225 – 0.414	4	Tetrahedral	ZnS
0.414 – 0.732	6	Octahedral	NaCl
0.732 – 1.0	8	Cubic	CsCl

Table 6.3 Radius ratio

6.7 Imperfection in solids:

According to the law of nature nothing is perfect, and so crystals need not be perfect. They always found to have some defects in the arrangement of their constituent particles. These defects affect the physical and chemical properties of the solid and also play an important role in various processes. For example, a process called doping leads to a crystal

XII U6 Solid State - Jerald.indd 192 22-12-2022 19:52:50